Corvus Health: FDE Anchor Case

Student Brief

How this case works

Your group is the Forward Deployed Engineering (FDE) team assigned to Corvus Health Insurance for a 10-week engagement. You will stay with this one client for all five sessions, and each session's deliverable is the input to the next. Decisions you make in Session 2 will come back to help or hurt you later.

- **This brief is deliberately incomplete.** Real clients never hand you the full picture. The facts you need are held by Corvus stakeholders, played by your instructor and TAs. You get them by asking good questions.
- **Stakeholders disagree with each other.** Part of the job is working out whose view is right, and for what.
- **Things will change mid-session.** Expect new information, new constraints and new demands. How you respond counts as much as your original plan.
- **Keep a running decision log.** Record each major decision, the reason, and what you assumed. You will need it in Session 6.
- **Rotate roles every session:** engagement lead, client liaison (the only person who talks to stakeholders), builder, scribe, and sceptic (whose job is to argue against the group's plan).

Corvus Health Insurance

Corvus is a fictional mid-sized health insurer with about 3.2 million members, most covered through employer group policies. It works with a network of roughly 4,500 hospitals, and members can be treated in two ways:

- **Cashless:** the hospital asks Corvus for pre-authorisation before or during treatment, and Corvus pays the hospital directly.
- **Reimbursement:** the member pays the hospital, then submits a claim to Corvus to get the money back.

Corvus has grown fast through acquisitions over the last eight years. Its operations still run on a mix of old and new systems. Claims operations, in two processing centres, employs about 140 claims processors and 25 medical reviewers (doctors and nurses who check clinical necessity).

The market is competitive. Large employer clients are starting to choose insurers based on hospital and employee experience, and Corvus has lost two corporate accounts in the last year. The board has asked leadership for a visible AI-driven improvement within the next two quarters. Corvus has signed an enterprise licence with your company's AI platform, and your FDE team is the first deployment.

The engagement brief

This is the email your team received from the Corvus COO the week before kick-off. It is all you know going in.

> **From:** Priya Menon, Chief Operating Officer, Corvus Health Insurance
>
> **Subject:** AI for claims: kick-off next week
>
> Team, welcome aboard. Our claims process is too slow and too manual, and our hospital partners and corporate clients are telling us so. I want to use AI to fix claims. Ideally it should read the claim, check it against the policy and approve the straightforward ones automatically, so my people can focus on the complex cases.

> The board will want to see progress quickly. I'd like a working demo within the first month and something live by the end of the engagement.
>
> You'll have access to my leadership team and to our claims operations staff. IT will sort out data access. Looking forward to working together.

Engagement terms: 10 weeks, one FDE team, access to Corvus staff for interviews, and the AI platform licence. Any production deployment must pass Corvus's IT security review.

Who you can talk to

Your client liaison can request time with any of these five people. Each interview slot is 7 minutes in Session 2 and 5 minutes in later sessions, so plan your questions before you ask.

Stakeholder	Role	What you know about them
Priya Menon	Chief Operating Officer, executive sponsor	Wrote the brief. Owns the budget and reports to the board on this engagement.
David Okafor	Head of Claims Operations	Runs both processing centres and all 140 processors. Hasn't commented on the engagement yet.
Rahul Iyer	Head of IT and Data	Owns core systems, data access and the security review. Joined Corvus 14 months ago.
Dr. Sarah Lim	Chief Medical Officer	Leads the 25 medical reviewers. Signs off on clinical policy.
Meena Joshi	Senior Claims Processor, 11 years at Corvus	Handles cashless pre-authorisation requests. Trains new processors.

Session assignments

Each session you present your deliverable to the client, played by your instructor. Expect to be challenged the way a real client would challenge you.

Session 2: Discovery and scoping (engagement weeks 1–2)

Your task: Work out what Corvus actually needs, and agree a scope you can deliver in 10 weeks.

Deliverable: a one-page scope document covering:

1. The problem statement, in the client's language, with the evidence behind it
2. The use case you will build, and what is explicitly out of scope
3. Two or three success metrics, with today's baseline and a target
4. A stakeholder map: who supports this, who is worried, and why
5. Your top three risks and assumptions

Session 3: Rapid prototyping (weeks 3–4)

Your task: Build a working prototype of your scoped use case and demo it to Corvus.

Inputs you'll receive: sample documents and business rules that fit your scope.

Deliverable: a working prototype (any tool your instructor allows) and a 5-minute demo script that shows a real case running end to end. Include one case where it fails, and say what you'd do about it.

Session 4: Integrating with enterprise data (weeks 5–6)

Your task: Plan how the prototype connects to Corvus's real systems and data.

Inputs you'll receive: a system map, IT and security policies, and a data extract.

Deliverable: an integration plan covering:

1. An architecture diagram showing data flow from intake to the processor's screen
2. What data you need, where it lives, and how you'll get access
3. How you handle personal and health data
4. A phased rollout path (for example: offline test, shadow mode, then live with human review)
5. Integration risks and your mitigations

Session 5: Evaluations (weeks 7–8)

Your task: Prove the system is good enough to go live, and define what "good enough" means.

Inputs you'll receive: a labelled test set.

Deliverable: an evaluation report covering:

1. The metrics you chose and why. Which errors cost Corvus most?
2. Your test set design, including the edge cases you added
3. Results, broken down by segment rather than one overall number
4. Go-live thresholds, and what happens when the system is unsure
5. How you will monitor quality after launch

Session 6: Handing off to the client team (weeks 9–10)

Your task: Leave Corvus able to run, improve and own the system without you.

Inputs you'll receive: details of the internal team who will take over.

Deliverable: a handoff package covering:

1. Ownership: who owns what (a RACI chart)
2. A runbook: what to do when something breaks, and who to call
3. How prompts, rules or models get updated, tested and released
4. A training plan for the processors and reviewers who use it
5. A 30/60/90-day success review plan, and your recommended next use case
6. Your decision log from the whole engagement

Exhibits

Exhibit 1: Claims operations dashboard, last quarter

This is a slide from David Okafor's monthly operations review. Read it closely.

Measure	Cashless pre-authorisation	Reimbursement claims
Volume per day	about 1,800 requests	about 2,600 claims
Average turnaround	6.5 hours	11 days
Target turnaround	3 hours (hospital contracts)	15 days (internal)

Measure	Cashless pre-authorisation	Reimbursement claims
Sent back to hospital or member for more information	38%	22%
Escalated to a medical reviewer	14%	9%
Rejected	6%	8%

Exhibit 2: Systems in use (as listed on the IT intranet)

- **ClaimCore:** core claims and policy system, from an outside vendor, in use since 2011
- **DocVault:** document management system that stores every scanned and uploaded document
- **Intake gateway:** receives pre-authorisation requests from hospitals by email, fax and the hospital portal
- **Corvus data warehouse:** reporting and analytics

Exhibit 3: Recent complaints (summary from the partner relations team)

- A 600-bed hospital chain says pre-authorisation delays are holding up patient discharges, and has threatened to leave the Corvus network.
- A corporate client's HR head says employees are "stuck in the hospital waiting for insurance".
- Processors in the second centre report overtime most weekends since the last acquisition.